

PRACTICAL 01: PROCESS CREATION & SYSTEMS EXPLORATION

1. Problem Statement & Requirements

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that is:

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait()`.
5. Display the Process ID (PID) of both parent and child processes.

Task 2: Using Linux terminal commands (`uname`, `lscpu`, `lsblk`, `ps`, `top`), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

2. Program Execution & Output

```
rishi@RISHI-VP:~$ cd OSSP
-bash: cd: OSSP: No such file or directory
rishi@RISHI-VP:~$ mkdir OSSP
rishi@RISHI-VP:~$ cd OSSP
rishi@RISHI-VP:~/OSSP$ mkdir exec
rishi@RISHI-VP:~/OSSP$ cd exec
rishi@RISHI-VP:~/OSSP/exec$ nano program1.c
rishi@RISHI-VP:~/OSSP/exec$ gcc program1.c -o program1
rishi@RISHI-VP:~/OSSP/exec$ ./program1
Enter a Linux command: ls
Parent Process PID: 585
Child Process PID: 586
Parent PID (from child): 585
program1  program1.c
Child process completed.
rishi@RISHI-VP:~/OSSP/exec$ uname -a
Linux RISHI-VP 5.15.133.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18
21:54:43 UTC 2026 x86_64 GNU/Linux
rishi@RISHI-VP:~/OSSP/exec$
```

3. Hardware & OS Inspection Commands

```
rishi@RISHI-VP:~/OSSP/exec$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 8
On-line CPU(s) list:    0-7
```

```

Vendor ID: GenuineIntel
Model name: Intel(R) Core(TM) i5-8300H CPU @ 2.30GHz
CPU family: 6
Model: 158
Thread(s) per core: 2
Core(s) per socket: 4
Socket(s): 1
Stepping: 10
BogoMIPS: 4608.00
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
  L1d: 128 KiB (4 instances)
  L1i: 128 KiB (4 instances)
  L2: 1 MiB (4 instances)
  L3: 8 MiB (1 instance)
NUMA node(s): 1
NUMA node0 CPU(s): 0-7

```

```

rishi@RISHI-VP:~/OSSP/exec$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 0 disk
sdb 8:16 0 159.4M 0 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/

```

```

rishi@RISHI-VP:~/OSSP/exec$ ps
PID TTY TIME CMD
336 pts/0 00:00:00 bash
697 pts/0 00:00:00 ps

```

```

rishi@RISHI-VP:~/OSSP/exec$ top
top - 03:25:48 up 10 min, 1 user, load average: 0.00, 0.03, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7865.6 total, 7384.0 free, 506.0 used, 117.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7359.6 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24428  13460 11624 S   0.0   0.2   0:01.70 systemd
    2 root        20   0    3180   2212  2072 S   0.0   0.0   0:00.02 init-systemd(UB
    7 root        20   0    3180   2048  1940 S   0.0   0.0   0:00.00 init
   50 root        19  -1   50392  16788 13512 S   0.0   0.2   0:00.37 systemd-journal
   67 systemd+    20   0   22540  14596 11980 S   0.0   0.2   0:00.22 systemd-resolve
   94 root        20   0   35248  12244  9188 S   0.0   0.2   0:01.34 systemd-udevd
  119 root        20   0    2888   1952  1820 S   0.0   0.0   0:00.11 chronyd-starter
  120 root        20   0    4472   3088  2844 S   0.0   0.0   0:00.02 cron
  121 message+   20   0    8820   5328  4669 S   0.0   0.1   0:00.13 dbus-daemon
  158 root        20   0   44092  29768 14628 S   0.0   0.4   0:00.45 networkd-dispat
  165 root        20   0   18432   9420  8176 S   0.0   0.1   0:00.25 systemd-logind
  167 root        20   0 1792300  11960  7304 S   0.0   0.1   0:00.23 wsl-pro-service
  181 syslog     20   0  220548   5784  4624 S   0.0   0.1   0:00.16 rsyslogd
  234 _chrony     20   0   20420   6356  5576 S   0.0   0.1   0:00.05 chronyd
  238 _chrony     20   0   12092   2408  1760 S   0.0   0.0   0:00.01 chronyd
  279 root        20   0    5492   2744  2556 S   0.0   0.0   0:00.02agetty
  289 root        20   0  123456  32716 18112 S   0.0   0.4   0:00.36unattended-upgr

```

334	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
335	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.06	Relay(336)
336	rishi	20	0	4268	5536	3864	S	0.0	0.1	0:00.17	bash
337	root	20	0	8648	5628	4784	S	0.0	0.1	0:00.01	login
383	rishi	20	0	22748	13384	10932	S	0.0	0.2	0:00.26	systemd
385	rishi	20	0	22908	4104	2104	S	0.0	0.1	0:00.00	(sd-pam)
411	rishi	20	0	4136	5412	3860	S	0.0	0.1	0:00.05	bash
707	rishi	20	0	8184	6120	4000	R	0.0	0.1	0:00.14	top